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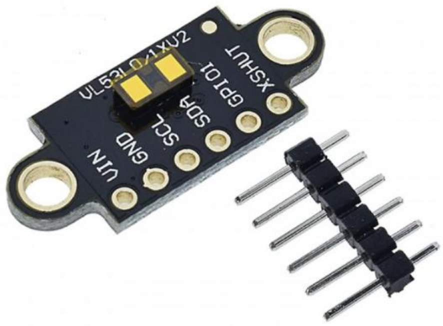
Concept Review
Time of Flight

Why Do We Need Time of Flight Sensors?

Many mechatronics systems rely on distance sensors for obstacle avoidance, object detection, and localization. Time of Flight sensors fall under the topology of distance measuring devices. Like the name suggests, Time of Flight sensors operate by sending out a pulse of light and measuring how long it takes to return to a sensor at which point a distance measurement can be estimated.

Time of Flight

Time of Flight (ToF) sensors employ both signal emission and signal detection to perform distance estimation. Light is emitted through a laser or an LED and is reflected by nearby objects with a time delay proportional to the distance of the object. By measuring the amount of time taken for the emitted signal to be detected back at the sensor (the time-of-flight), the distance of objects from the sensor can be estimated. ToF sensors can come in varying sizes and configurations, and two different ones are shown in Figure 1.



a. Tiny Electronics VL53L1X

b. Adafruit TFmini

Figure 1: Different configurations for laser-based Time of Flight sensors.

Figure 2 shows an example of a laser-based time of flight sensor which estimates the straight-line distance to a subject of interest.

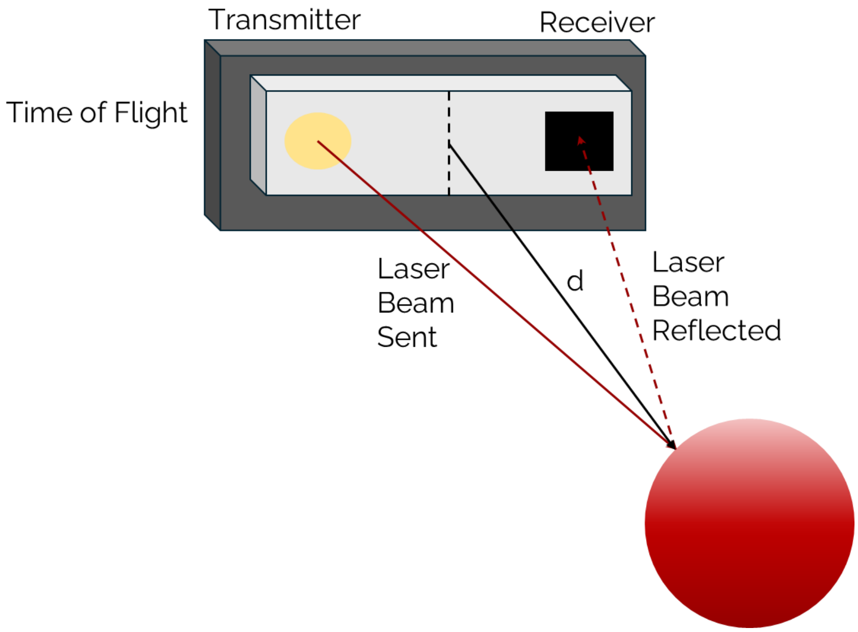


Figure 2: Distance estimate from a laser-based time of flight sensor

Using the time delay from when light is emitted to when it is received, Δt, and the speed of light, c, we can use the following equation to estimate the distance, d, to an object:

$$d = c\Delta t/2 \tag{1}$$

In equation 1, the speed of light c is a constant equal to 299,792,458 m/s. Depending on the sensor manufacturer, this calculation may be done within the sensor directly. As an example, the STM VL53L8CHV0GC will provide distance values directly.

Another interesting characteristic of modern Time-of-Flight sensors is multizone measurements. Multizone refers to a grid like series of measurements which provide more information than a single beam. This information can be useful if you're interested in interpreting the shape of an object within proximity to the Time-of-Flight sensor. Sensors such as the STM VL53L8CHV0GC can provide you with multizone distance values.

Time of Flight sensors also have different technical characteristics based on the manufacturer's specifications. Examples of different Time of Flight sensor technologies:

- VL53L8CHV0GC uses a single photon avalanche diode (SPAD) detection array.
- ADSD3100 uses a Complementary Metal Oxide Semiconductor (CMOS) sensor.

For a VL53L8CHV0GC the SPAD detector array can generate a multizone distance measurement as seen in Figure 3. For this example, a maximum distance of 0.5m is represented by the dark red and a bright blue are values close to the sensor array.

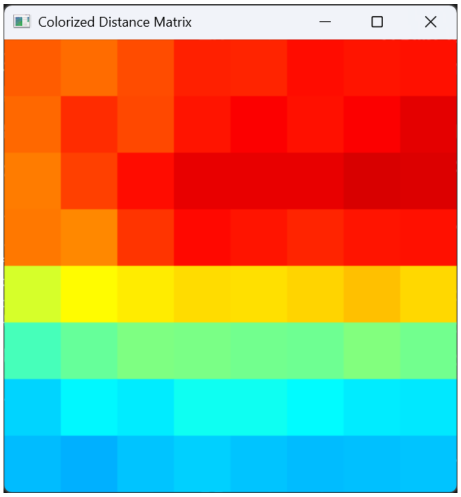


Figure 3: Colorized multizone distance information from VL53L8CHV0GC

Selecting a Time-of-Flight Sensor

Datasheets are used to select an appropriate sensor for a specific application and to determine additional design constraints on the rest of the system. For time-of-flight sensors, some key datasheet considerations include the following:

Dynamic range: This describes the full range of operating conditions, including both distance and reflectivity, that the sensor is expected to detect. This value is the ratio of the maximum to the minimum values of each measurement parameter.

Resolution: This refers to the smallest detectable change in distance. Lower resolutions may be sufficient for less critical tasks, while high resolutions are useful for applications that require more detailed measurement, such as imaging.

Field of view: This describes the dimensions of the area within which objects are detectable by the sensor. In sensors containing both an emitter and a receiver, the field of view is determined by the field of view of both components.

Sampling rate: This value indicates how quickly the sensor can output measurements. The minimum sampling rate depends on the response time requirements for the specific application.

Comparison to Other Distance Sensors

The ToF sensor falls within a category of sensors that employ various technologies for distance measurement. The type of sensor that is best suited for an application depends on specific requirements such as cost, range, resolution, accuracy, and speed. Compared to an ultrasonic sensor, which uses sound waves instead of light to measure distance, ToF sensors provide benefits such as higher reading frequency, resolution, and accuracy but at a generally greater cost. The accuracy of a ToF sensor can vary depending on the reflectivity of the target surface, which can vary by material and color.

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